

Intel Core Ultra 9 285K

A New Beginning

Intel Arrow Lake processors mark a significant step forward for the major chip

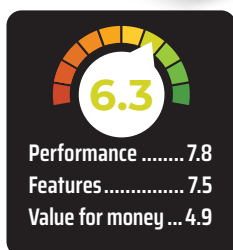
manufacturer which has been in the news quite often of late. The new Intel Core Ultra 200S series processors are the first to include silicon made from TSMC on the package, however, this is isn't the first time that Intel is using TSMC made silicon. TSMC has been making chipsets for Intel motherboards for ages and with the Arrow Lake launch, that relationship has only gotten stronger. So what does that mean for the consumer? Better performance? More power-efficiency? Let's find out in today's review of the Intel Core Ultra 9 285K and Ultra 5 245K.

INTRODUCING, ARROW LAKE-S

Intel recently unveiled their latest mobile processors in the form of Core Ultra Series 2 a.k.a. Lunar Lake boasting of several new improvements especially on the AI acceleration front along with all new CPU cores and several efficiency gains with regards to power consumption and thermals.

Arrow Lake borrows a lot of things from Lunar Lake including the two new core architectures – Lion Cove for the P-Core (Performance Cores) and Skymont for the E-Core (Efficiency Cores). The P-Cores also get a larger L2 Cache and the E-Cores get access to the L3 Cache. The NPU (Neural Processing Unit) isn't as powerful as Lunar Lake with just 13 TOPS compared to the 48 TOPS that Lunar Lake's NPU4 is capable of. This is the first ever NPU on an Intel Desktop processor and there are plenty of other avenues for AI acceleration on desktop

₹58,900



systems so more of the real estate on the CPU package can be dedicated to CPU compute rather than AI acceleration. In Intel's terms, this would be 2x Gen3 NPU cores. The mobile chips get the newer Gen4 NPU Cores. The iGPU (Integrated Graphics Processing Unit) is also new with about 320 Execution Units. Aside from these improvements, there's integrated Thunderbolt 4 and Wi-Fi 6E support. Though it must be pointed out that most top-tier motherboards have already integrated the newer Wi-Fi 7 standard onto their motherboards.

The new Intel Core Ultra 200S processors are expected to reduce power consumption across a wide assortment of workloads and even provide higher multi-threaded performance over Raptor Lake-R processors in some work-

loads. One of the major changes to the physical design is that the Arrow Lake processors are the first to have a disaggregated design which is a different way of saying that they use a chiplet design much like what AMD does with their Ryzen processors. However, the interconnect technologies that AMD and Intel use are quite different so it's not exactly the same. That being said, a disaggregated structure allows Intel to pick tiles with different performance metrics across different SKUs. There's a lot more to dig through on the architecture front but we'll stick to the review for the Core Ultra 9 285K for today.

SPECIFICATIONS

We've already reported on the launch of the Intel Core Ultra 200S series with this first wave of processors having just five SKUs but they're all heavy hitters. At the top of the stack is the Intel Core Ultra 9 285K which has 8 P-Cores and 16 E-Cores for a total of 24 Threads. Notice something? There's no multi-threading! Intel has done away with HyperThreading and has introduced cores that just do one thread each. What this entails for heavily-threaded applications will be seen shortly. The flagship 285K is followed by the Ultra 7 265K and 265KF and then we have the Ultra 5 245K and 245KF. The Ultra 5 245K comes with 6 P-Cores and 8 E-Cores.

The Intel Xe-LPG iGPU on these processors sport 4 GPU Cores and the NPU can do 13 TOPS as we'd mentioned previously. The Intel Core Ultra 9 285K has 36 MB of L3 Cache on the compute tile which can be accessed by all the cores. Moreover, unlike previous designs where the E-Cores would be on one side of the silicon and P-Cores would be on the other, the Arrow Lake-S processors now have them placed in a sandwiched manner. So the P-Cores and E-Core clusters are interleaved.

The E-Cores come with a base clock of 3.2 GHz and go up to 4.6 GHz whereas the P-Cores start off at 3.7 GHz and